



Certificate of Analysis

ILS Laboratories

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TB-500 - 10mg

Tested for:

PASS

COA #:

Lot Number:

Accession #:

Labeled Content: 10mg

Analysis Date:

08/17/2026

Appearance:

Good

Sample Matrix:

Lyophilized

Date Received:

08/13/2026

Method: Full QC Panel

Identity

TB-500

Peptide Purity

99.58%

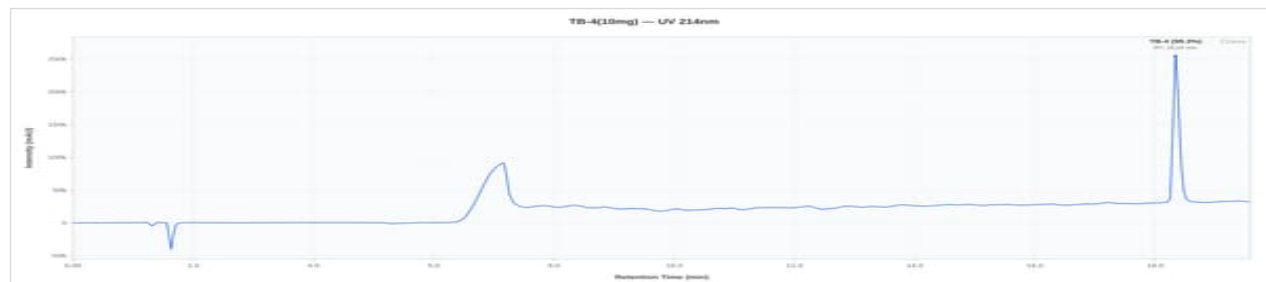


Fentanyl Free

Full QC Panel

Analyte	Specification	Result	Unit	Status
Peptide Purity (HPLC)	$\geq 95.0\%$	99.58%	%	PASS
Net Peptide Content	Report Only	10.42	mg	N/A
Identity (HPLC-RTM)	TB-4	Confirmed	-	PASS
Fentanyl Screen	Immunoassay, 50 ng/mL cutoff	Not Detected	-	PASS

HPLC Chromatogram



TB-500 10mg - TB17010: UV Chromatogram

Heavy Metals Analysis (ICP-MS)

Test	Specification	Result	Status
Arsenic (As)	NMT 1.5 ppm	Not Detected	PASS
Cadmium (Cd)	NMT 0.5 ppm	Not Detected	PASS
Lead (Pb)	NMT 1.0 ppm	Not Detected	PASS
Mercury (Hg)	NMT 1.5 ppm	Not Detected	PASS
Chromium (Cr)	NMT 10.0 ppm	Not Detected	PASS

Sterility Testing (PCR)

Test	Specification	Result	Status
Sterility (PCR)	No Growth	No Growth	PASS

Endotoxin Testing (USP <85>)

Test	Specification	Result	Status
Endotoxin (USP <85>)	Report Result	0.084 EU/mL	Reported

About this result: Endotoxin is reported as a quantitative value. Acceptable limits vary by product type and matrix, so no universal pass/fail threshold applies to RUO products. This result is below commonly referenced endotoxin thresholds.

Notes & Methodology

1. Date Tested: 08/17/2026. Methods: Full QC Panel.
2. The sample was confirmed to be TB-500 by HPLC. Identification by chromatographic retention time comparison with a reference standard.
3. Elemental impurities analyzed by ICP-MS per USP <233> methodology. Acceptance criteria are internal laboratory quality screening limits for research-use materials and do not represent evaluation against any specific pharmacopeial monograph or product specification.
4. Endotoxin tested per USP <85> kinetic chromogenic method. Acceptance criteria per client specification.
5. Peptide purity determined by RP-HPLC area normalization at 214 nm. Value represents the percentage of the target peptide relative to all peptide-related peaks. Non-peptide process-related impurities, if detected, are excluded from the calculation.
6. Accreditation Scope: ILS Laboratories maintains ISO/IEC 17025 accreditation for activities listed in its published PJLA scope. Peptide testing reported on this certificate is performed for Research Use Only.
7. This certificate reports analytical results for the sample(s) submitted to ILS Laboratories for testing. All sample identifications, product names, brand names, lot/batch numbers, and labeled content are provided by the submitting party and have not been independently verified. Results reflect only the analytical testing of the specific sample(s) provided and do not constitute an evaluation or endorsement of the manufacturer's processes, quality systems, or overall production. ILS Laboratories makes no representations regarding the intended use, safety, efficacy, regulatory status, or intellectual property ownership of any materials tested.



Issued by ILS Laboratories
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8/17/2026